

BUSINESS ANALYST CASE STUDY • SELF-INITIATED PORTFOLIO PROJECT

ZestEats Subscription Retention Optimization

Reducing ZestPrime subscriber churn through prediction-led, personalized retention interventions.

Prepared by Satyam Dubey

Fictional company, scenario & data — created for learning and portfolio purposes

Disclaimer & Why I Built This

This is a self-initiated Business Analyst case study created for learning and portfolio purposes. The company, business scenario, and data used are fictional but designed to reflect realistic business analysis practices.

Why I built this

I wanted to practice the full BA toolkit on something close to a real product problem rather than a generic template. Subscription churn felt like a good fit — it has real ambiguity, competing stakeholder interests, and a mix of behavioral and technical causes. I worked through it the way I'd run a real engagement: discovery, a BRD/FRD, prioritization trade-offs, SQL analysis, and an honest delivery plan.

- 1 Stakeholder discovery**
Workshops across Product, Data, Engg & CX
- 2 BRD & FRD writing**
Business rules, scope, and key user flows
- 3 SQL analysis**
Churn, cohort & payment-recovery queries
- 4 Prioritization**
RICE, MoSCoW, and MVP scoping
- 5 Agile delivery**
Backlog, sprint plan, stakeholder management

Executive Summary

ZestEats is a fictional food-delivery platform whose subscription product, ZestPrime, was losing roughly 1 in 7 subscribers every month — a churn rate of around 14%. That was eating into subscription revenue and pressuring unit economics, since subscribers order far more often than non-subscribers.

I worked through this as a Business Analyst, collaborating with simulated Product, Data Science, Engineering, and CX stakeholders to scope a Subscription Retention System — combining churn prediction, personalized intervention offers, and a payment-recovery flow.

My approach: start with discovery, document requirements multiple teams could commit to, prioritize given limited engineering time, and plan delivery honestly — including what's done, in progress, and still backlog.

~14% → <8%

Target monthly churn reduction

₹1–1.5 Cr/mo

Estimated recovered subscription revenue

>80% AUC-ROC

Churn-prediction model accuracy target

**All figures are illustrative estimates for this case study, not measured outcomes.*

Company & Product Background

ZestEats

A fictional food-delivery marketplace operating across a number of Indian cities. Like most platforms in this space, it earns money two ways: a commission on restaurant orders, and a paid subscription tier.

ZestPrime

ZestEats' subscription tier (~₹99/month or ₹799/year). Subscribers order more often and spend more per order than non-subscribers — which is why retaining them matters more than the subscription price alone suggests.

Assumed scenario: a subscriber base in the low hundreds of thousands — large enough that churn has real revenue impact, small enough to reason about without enterprise-scale infrastructure assumptions.

ZestPrime benefits

Free delivery above a minimum order value

Flat cashback on every order

Member-only restaurant deals

Faster delivery SLA

Ad-free app experience

The Business Problem

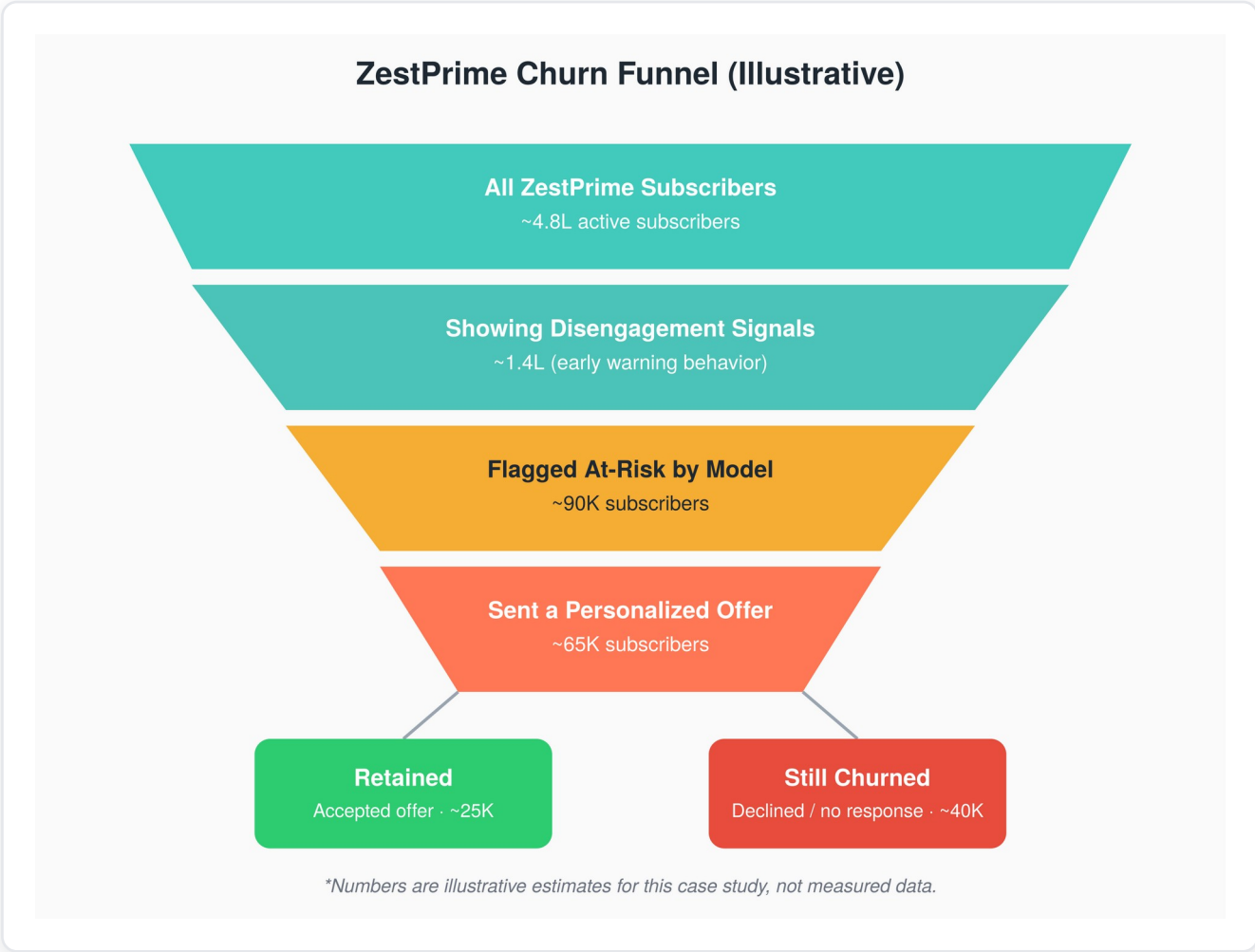
- 1

~14% monthly churn
On the ZestPrime subscriber base
- 2

Voluntary churn (~62%)
Value erosion, low usage, competitor switch
- 3

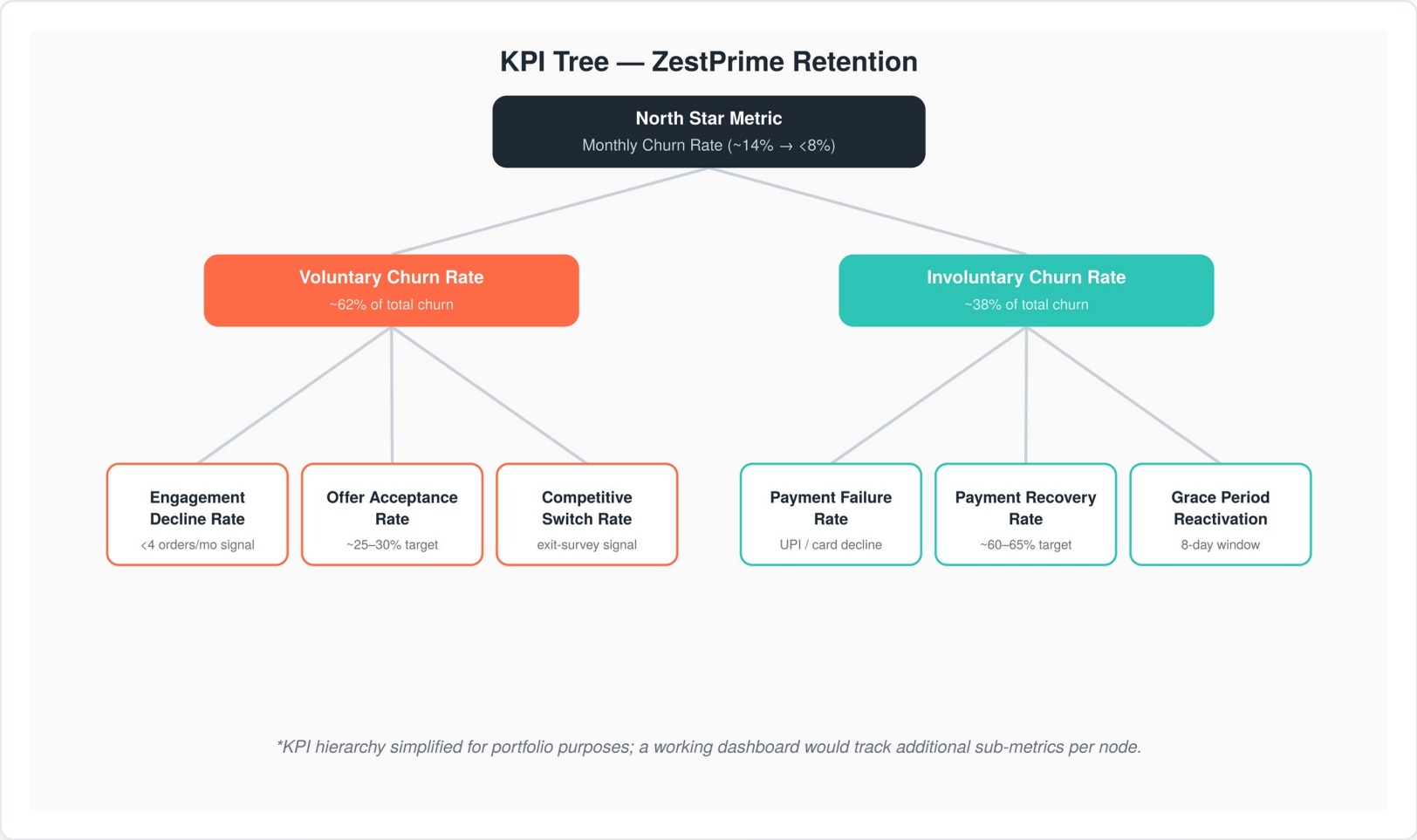
Involuntary churn (~38%)
Payment failures with no retry logic
- 4

Acquisition isn't a fix
CAC is far higher than retention cost



Goals & Success Metrics

Primary goal: bring monthly churn from ~14% to under 8% within ~6 months, recovering a meaningful slice of subscription revenue. The tree below breaks that North Star down into the KPIs that actually drive it.



SECTION 05

Discovery — Stakeholder Voices

“Success in six months looks like churn under 8% — that's the commitment I'd want to make upward.”

VP Growth & Retention

“We already track a lot of behavioral signals. We just haven't built anything that acts on it for retention.”

Head of Data Science

“Today this is entirely manual — a save team calling people who've already cancelled. It doesn't scale.”

CX Lead

“Whatever gets built needs to plug in through APIs and events — not a rewrite of the billing service.”

Engineering Lead

Business Requirements (BRD Highlights)

1

BR-1 — Early risk identification

Flag at-risk subscribers ~30+ days before they'd lapse

2

BR-2 — Personalized intervention

Match offer to likely churn reason, not a blanket coupon

3

BR-3 — Real payment recovery

Grace period + retries before cancelling on a failed payment

4

BR-4 — Leadership visibility

Shared churn & intervention reporting for Finance/Product/CX

Key business rules

Max ~2 retention messages/week per subscriber, across all channels

45-day cooldown after an accepted offer before targeting again

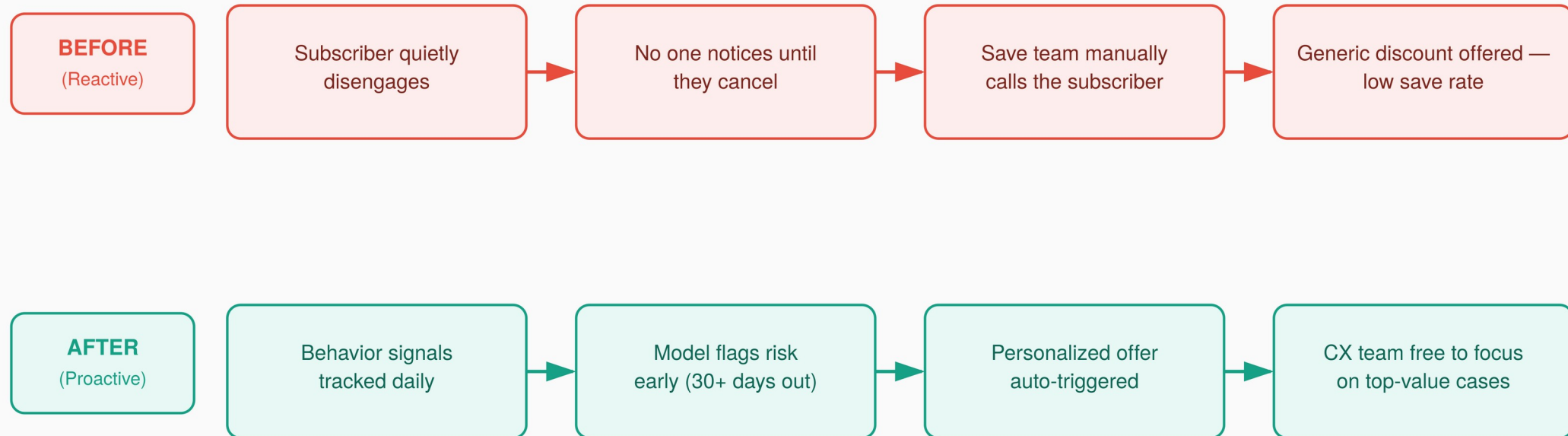
Discount-style offers capped as a % of subscription fee

Top-tier (VIP) subscribers routed to a human agent, not automation

FRD — Before vs. After the Retention System

Three simplified system pieces: a risk-scoring signal, an offer/decision layer that respects the business rules, and a payment-recovery flow. Kept light on architecture detail intentionally — that's Engineering's scope, not the BA's.

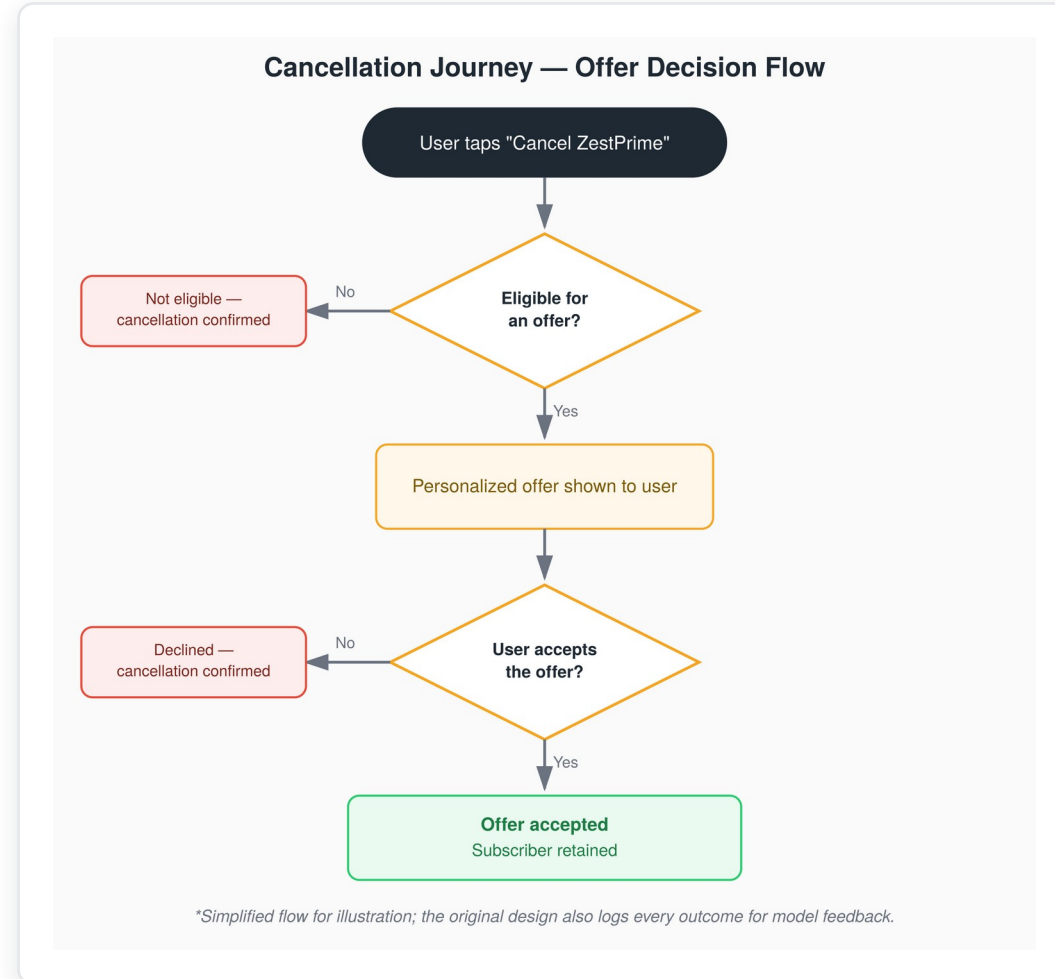
Retention Process — Before vs. After (Illustrative)



**Simplified for illustration; an actual delivery plan would include additional steps and edge cases.*

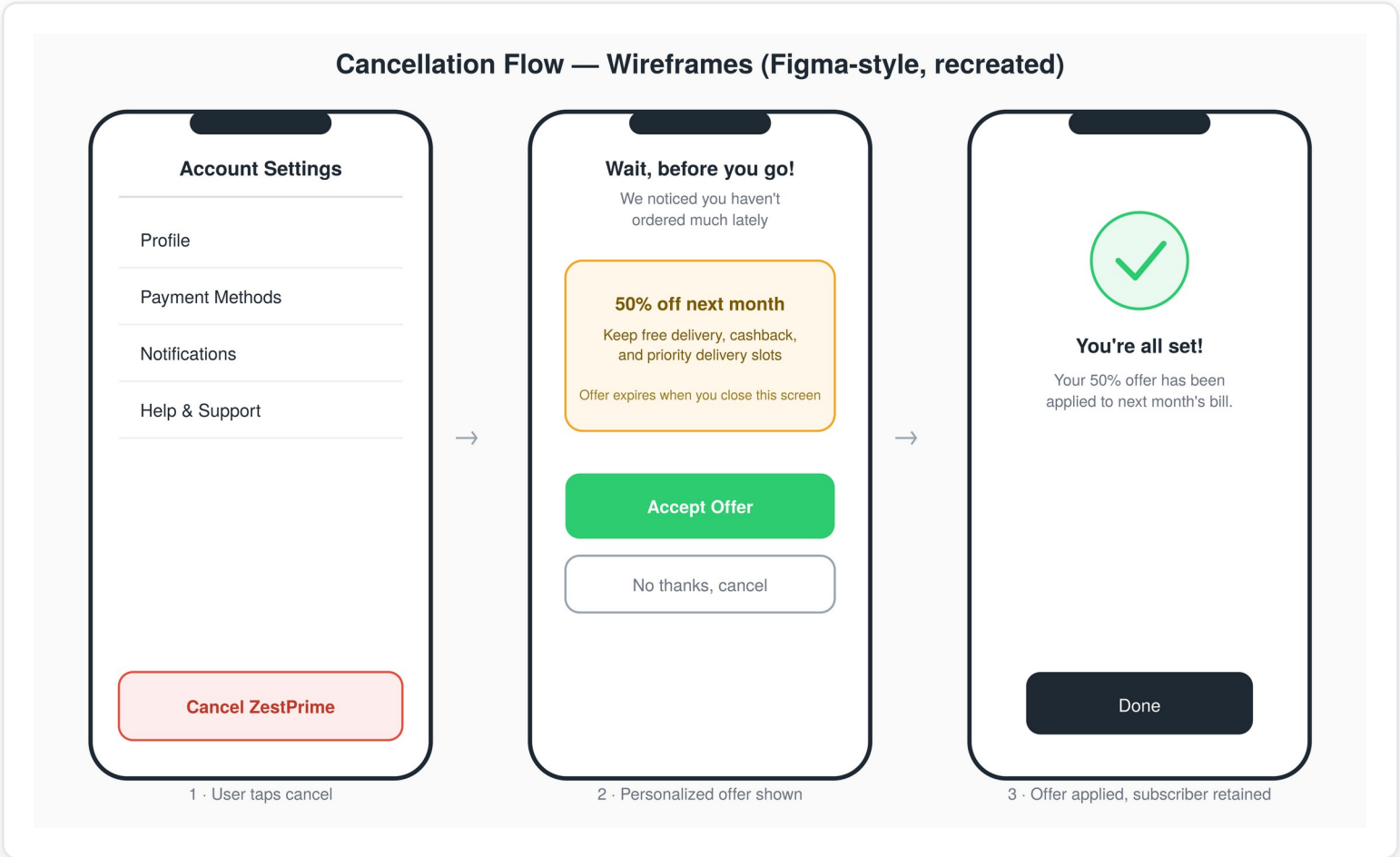
Key User Flow — Cancellation Moment

I prioritized this flow above almost everything else: it's the highest-intent moment in the churn journey — the user is actively telling you they're leaving, right now.



Cancellation Flow — Wireframes

Recreated from the original Figma-style mockups referenced in the FRD: settings → personalized offer → confirmation.



Agile User Stories (Sample)

ID	Story	Pts
US-01	As a subscriber, I want to see a relevant offer when I try to cancel, so I don't give up a benefit I'd actually use if reminded.	8
US-02	As the system, I want to check cooldown status before generating an offer, so we don't repeatedly target someone who recently said no.	3
US-03	As a subscriber whose payment failed, I want a grace period and a chance to fix payment, so I don't lose access over a fixable issue.	5
US-04	As a CX lead, I want top-tier subscribers auto-routed to a human agent, so our highest-value relationships get a personal touch.	3
US-05	As a Product lead, I want a simple dashboard of churn & offer performance, so I don't request a one-off report every week.	5
US-06	As the system, I want to log every intervention outcome, so the team can review what's actually working each month.	2

Full backlog in a real project would run 15–20+ stories across these themes; shown here as a representative sample.

SECTION 10

SQL & Data Analysis

Before jumping to "build a model," I wrote a small set of SQL queries against a simulated subscriber dataset to understand the problem.

1

Churn rate by engagement level

Confirmed <4 orders/month strongly predicts lapsing within ~60 days

2

Cohort retention over time

Tracked how signup cohorts retain across their first 12 months

3

Payment recovery by instrument

Compared UPI vs. card retry success to order the retry sequence

4

High-risk subscriber pull

Generated "today's list" for the CRM intervention queue

Why this mattered

The payment-instrument analysis is a good example of SQL changing a decision, not just decorating a slide. If one payment method recovers far better on the first retry, that should directly change the retry sequence order — not just get mentioned as an interesting finding.

Prioritization — RICE, MoSCoW & MVP

MUST

- Cancellation-moment offer flow
- Payment recovery / grace period
- Basic risk-scoring capability

SHOULD / COULD

- Multi-channel orchestration (WhatsApp, phone)
- Deeper analytics dashboard drill-downs
- Win-back campaign for already-churned users

WON'T (this phase)

- Full referral / growth feature
- Automated win-back for lapsed subscribers

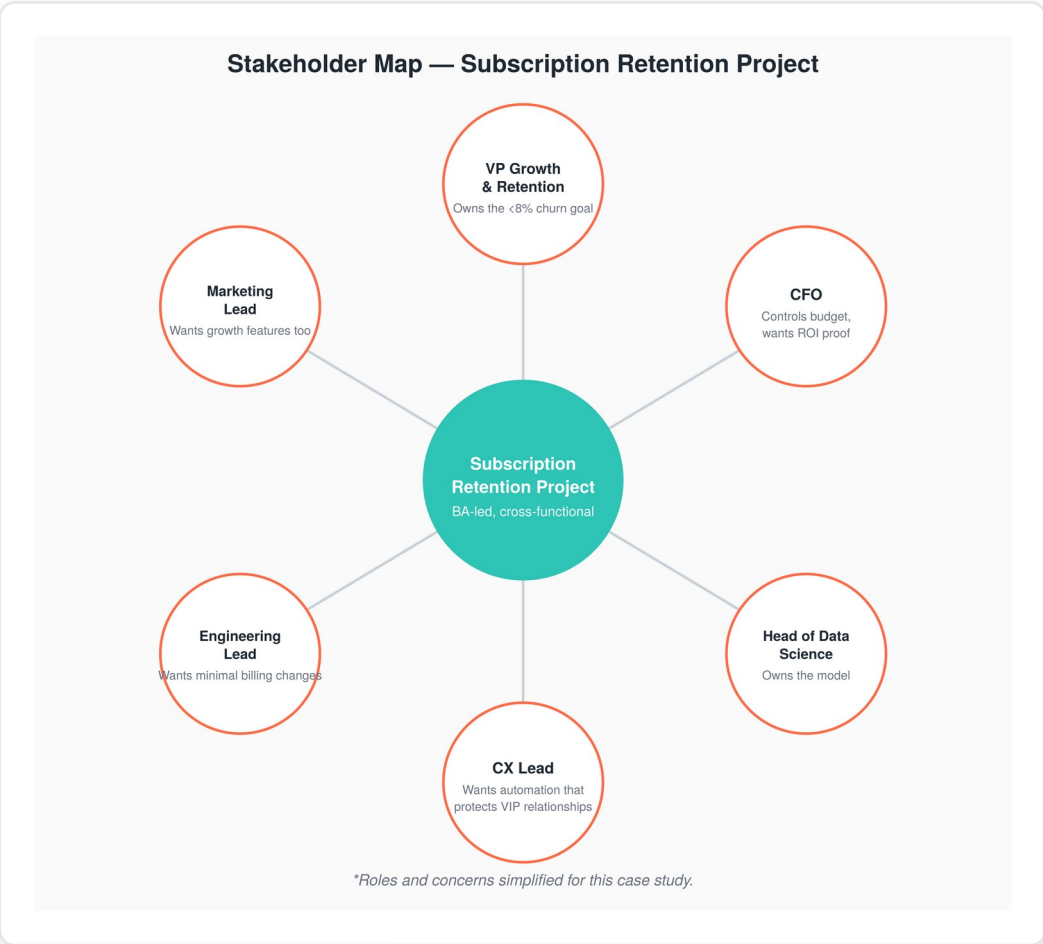
MVP definition: Churn risk scoring + cancellation interstitial + payment recovery. Chosen because together they address the largest share of the problem with the least dependency on each other.

Agile Delivery Plan

Epic	Description	Status
Churn risk scoring (v1)	Basic risk score + reason tagging for active subscribers	Completed
Cancellation offer flow	Offer shown at the cancellation moment, respecting business rules	Completed
Payment recovery flow	Grace period + retry sequence for failed renewals	In Progress
Retention reporting dashboard	Churn trend, offer performance & recovery views	In Progress
VIP routing to human agents	Auto-routing top-tier subscribers to a CX specialist queue	Planned
Referral banner (Marketing ask)	In-app referral promotion, raised mid-project	Backlog

Statuses are deliberately mixed — not every epic in a real project is "done," and pretending otherwise would undercut the point of this case study.

Stakeholder Management



Scope-creep scenario

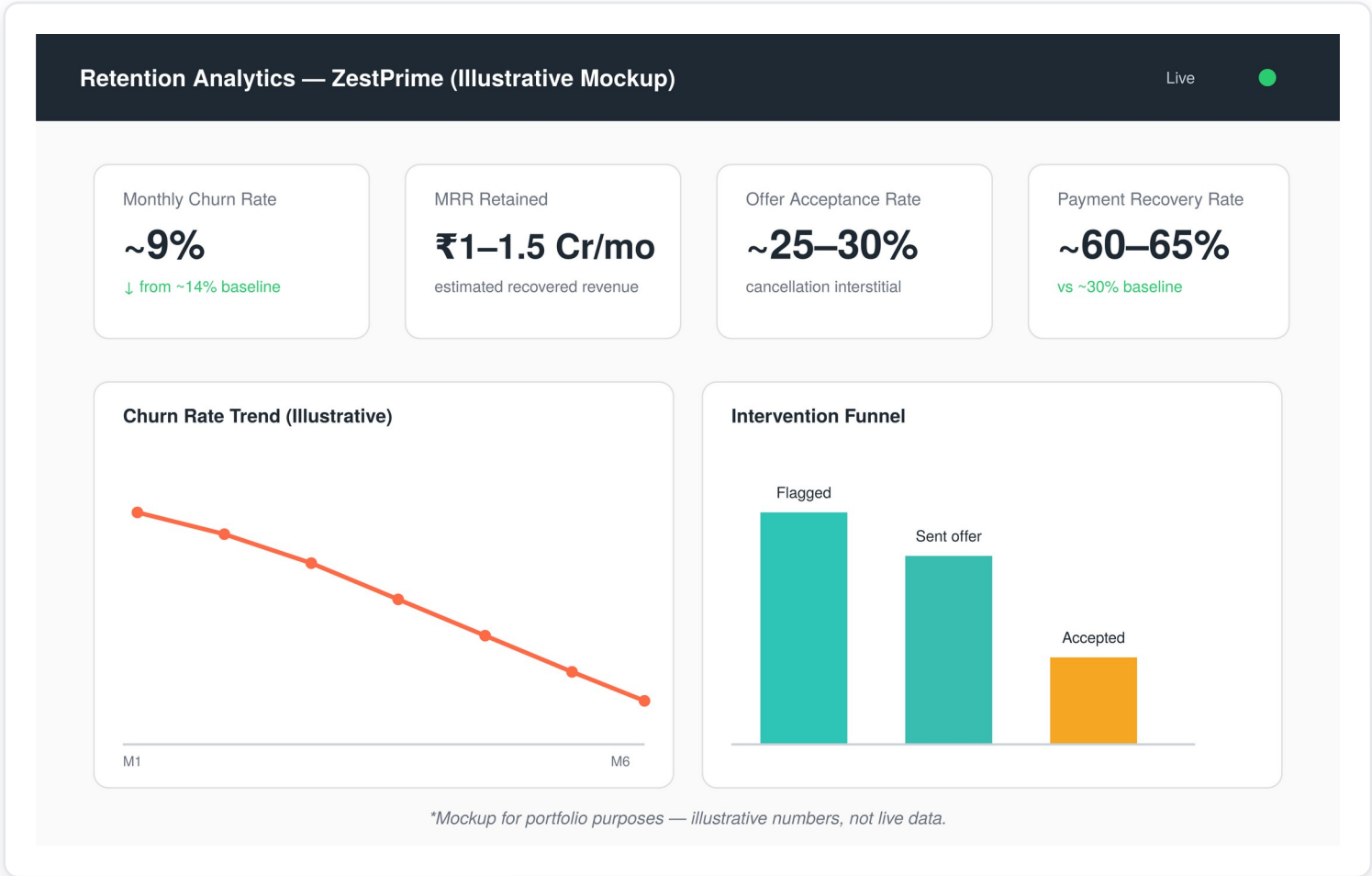
Marketing asked to add an in-app referral banner mid-sprint, which threatened to delay the highest-priority cancellation-offer flow.

- Acknowledged the idea's merit instead of dismissing it
- Roughly quantified the cost of delaying the core flow
- Proposed Release 2 with a committed timeline
- Took both options to the sponsor for a data-backed call

Outcome: Scope held. Referral feature got a committed Release-2 slot — Marketing felt heard, no relationship damage.

Results (Illustrative)

Modeled results if the plan played out as designed — not measured outcomes. In a real project, these would come from an A/B test or holdout comparison.



Lessons Learned & Reflections

1

Data should reorder priorities, not just justify them

Finding that ~38% of churn was payment-related (not a decision at all) changed what I'd build first.

2

The highest-intent moment is the best intervention point

Catching someone mid-cancellation beats predicting and emailing them three weeks earlier.

3

Visual requirements reduce ambiguity

A rough wireframe of the cancellation screen made offer-logic requirements far easier to agree on.

4

Stakeholder management is proactive, not reactive

The scope-creep conversation went smoothly because it used data and options, not a negotiation trick.

LIMITATIONS & ASSUMPTIONS

- Company, data, and metrics are entirely fictional — illustrative, not measured
- Simulated business environment — discovery sessions & conflicts are constructed, not transcripts
- Architecture is intentionally simplified — full engineering detail sits outside BA scope
- Role scoped to Business Analyst only — collaborating with Product, Engineering, Data & Ops, not personally building the model or dashboard

Thank you

I'm happy to walk through any section in more depth — particularly the requirements decisions and prioritization trade-offs, which is where I'd want a BA conversation to focus anyway.

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Full case study, SQL, diagrams & source files available on GitHub